Grammeme Order in the Task of Neural Morphological Tagging

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Abstract

This research investigates whether the order in which grammemes are predicted affects the accuracy of neural morphological tagging for Russian. We reimplemented the SEQ model [Tkachenko and Sirts, 2018] and evaluated several grammeme ordering strategies on multiple corpora. Our results show that prediction order has a statistically significant effect on tagging quality, with differences concentrated in a small set of morphologically challenging grammemes such as case, animacy, and part of speech. Project code: `https://github.com/nktshch/morph-orders`.

1 Introduction

Morphological tagging assigns a morphological tag to each word in a sentence, where a tag is composed of grammemes encoding categories such as case, gender, and number. It is a key step in NLP pipelines, used for annotated corpora [Savchuk et al., 2024], named entity recognition [Güngör et al., 2018], text classification [Mounica and Lavanya, 2024], and machine translation [Nzeyimana, 2024].

Neural network models now dominate morphological tagging [Baxi and Bhatt, 2024], with the most accurate approaches generating tags as composite sequences of grammemes [Can et al., 2022, Tkachenko and Sirts, 2018]. The SEQ model [Tkachenko and Sirts, 2018] generates grammemes sequentially, each conditioned on the previous predictions. [Tkachenko and Sirts, 2018] is the first paper to show that sequential generation yields the best accuracy. However, the impact of grammeme prediction order has not been studied before. This research fills that gap by comparing different ordering strategies and showing, both quantitatively and qualitatively, that order affects tagging accuracy.

1.1 Team

Nikita Shcherbak implemented the model and conducted experiments.
Ivan Konyushenko performed qualitative analysis and contributed to the writing.

2 Related Work

Order has been shown to affect quality in sequence generation tasks [Vinyals et al., 2016, Yang et al., 2019]. In multi-label classification, matching human annotation order improves results [Nwana and Chen, 2016]. In morphological inflection, ordering morphological segments improves generalization [Shcherbakov and Vylomova, 2023]. However, the effect of prediction order in morphological *tagging* has not been studied.

Grammemes are known to interact linguistically [Xrakovskij and Malchukov, 2016], and morphological features appear to be processed sequentially rather than as an unordered set [Bock and Miller, 1991, Schreuder and Baayen, 1995]. Different studies use different grammeme orderings: alphabetical (as in Universal Dependencies [Nivre et al., 2020]), frequency-based [Oh et al., 2019], or corpus-defined [Boguslavsky et al., 2024]. No prior work has systematically compared these orderings for tagging accuracy.

3 Model Description

The SEQ model [Tkachenko and Sirts, 2018] uses a bidirectional LSTM encoder and an LSTM decoder that generates grammemes one at a time. The probability of a tag $t_i = f_{i1}, \ldots, f_{ik}$ for token w_i is:

$$p\left(t_i | w_i\right) = \prod_{j=1}^{k} p\left(f_j | f_0, \ldots, f_{j-1}, w_i\right),$$

where f_0 is a special start-of-sequence grammeme. At each decoding step, the hidden state is updated and a linear transformation over the grammeme vocabulary produces the next prediction. The architecture is illustrated in Figure 1.

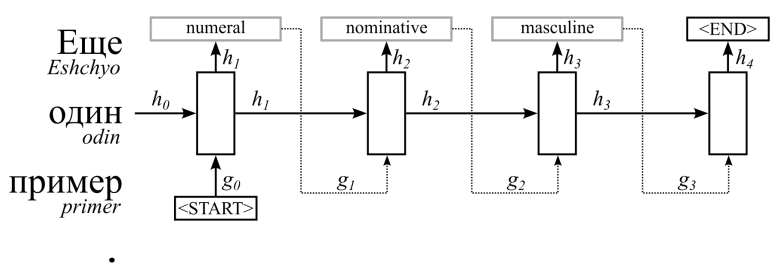

Figure 1: The process of generating grammemes in the SEQ model.

We reimplemented the model in PyTorch [Paszke et al., 2019], preserving the original logic and hyperparameters. We varied only the grammeme prediction order, which is controlled by a single hyperparameter.

3.1 Orders of Prediction

When selecting orders to consider in our work, we refered to the existing literature on morphological tagging. We considered the following orders:

- the order in which grammemes are placed in the corpus, which we will henceforth call the standard order, since the authors of SEQ used it. Using standard order is common practice since it doesn't require changing data in any way.
- the order corresponding to decreasing frequency of grammeme occurrence in the training set, since it was shown that prediction accuracy depends on frequency in similar tasks [Yang et al., 2019]. This order was also used in other works on morphological tagging [Oh et al., 2019]. We denote it as "Grammemes↓".
- Besides the standard and "Grammemes↓" orders, we considered additional orders, including the corresponding reverse variants. The order corresponding to increasing grammeme frequency will be denoted "Grammemes↑".
- The same orders but with part of speech being in the first position. For such orders, the name has "POS" added at the beginning, e.g., "POS, Grammemes↑". We also added orders in which morphological categories (rather than individual grammemes) are sorted by frequency. Orders using categories for frequency sorting include the word "Categories" in their name.

The choice of order is controlled by a separate hyperparameter. It determines the order in which grammemes are fed to the model during training, as well as what position each grammeme of a token takes when computing the loss function.

4 Dataset

We evaluated on three Russian corpora. GSD and SynTagRus from Universal Dependencies versions 2.1 and 2.15 [Boguslavsky et al., 2024, Nivre et al., 2020] were used to verify our implementation against the original SEQ results. Version 2.15 includes corrected annotations and an updated morphological standard [Droganova et al., 2018].

All corpora are publicly available for research purposes.

5 Experiments

5.1 Metrics

Each model was trained 5 times per corpus and order. The primary metric is the micro-averaged F-score over grammemes:

$$F\text{-}score_{micro} = \frac{\sum_{c=1}^{C} 2 \cdot TP_c}{\sum_{c=1}^{C} (2 \cdot TP_c + FP_c + FN_c)},$$

where TP_c, FP_c and FN_c are the numbers of true positives, false positives, and false negatives for grammeme c.

Per-grammeme F-scores are also computed to identify where ordering effects are most pronounced. Accuracy (fraction of fully correct tags) is reported for comparison with the original SEQ results.

5.2 Experiment Setup

For each corpus–order combination, means and standard deviations are computed across 5 runs. Best and worst orders per corpus are identified, and grammemes with statistically significant F-score differences between them are selected for qualitative analysis.

5.3 Baselines

The standard (corpus-defined) order serves as the baseline, consistent with the original SEQ paper.

6 Results

Table 1 shows micro-averaged F-score for each corpus and order. Bold values mark the best result per corpus; underlined values differ statistically significantly from the standard order.

Order	**GSD 2.1**	**SynTagRus 2.1**	**GSD 2.15**	**SynTagRus 2.15**
Standard	96.23 ± 0.08	98.54 ± 0.03	96.38 ± 0.07	97.44 ± 0.04
Reverse	**96.24 ± 0.06**	98.54 ± 0.01	96.33 ± 0.05	97.46 ± 0.04
"Grammemes↓"	96.14 ± 0.06	98.54 ± 0.02	<u>96.20 ± 0.06</u>	**97.47 ± 0.03**
"Grammemes↑"	<u>96.12 ± 0.04</u>	98.53 ± 0.03	96.34 ± 0.06	97.46 ± 0.03
"POS, grammemes↑"	96.18 ± 0.07	**98.57 ± 0.02**	96.31 ± 0.04	97.46 ± 0.02

Table 1: Mean micro-averaged F-score (%) per corpus and grammeme prediction order.

The difference between the best and the worst orders is statistically significant for all corpora–metric pairs except SynTagRus 2.15. The most affected grammemes per corpus are shown in Tables 2–3.

Qualitative analysis reveals that the performance differences trace back to grammeme dependencies. For instance, on GSD, the "POS, Grammemes↑" order correctly tagged the adjective *хозяйственные* 'household' with nominative case in a sentence *«Отличие в том, что хозяйственные объекты, которые*

Grammeme	F-score (Reverse)	F-score (“Grammemes↑”)	Mistake count
Animacy=Inan	97.23	97.10	143
POS=ADJ	95.98	95.68	45
Aspect=Imp	96.69	96.33	33
POS=DET	95.00	95.76	15

Table 2: Grammemes with statistically significant difference between best and worst orders on GSD 2.1.

Grammeme	F-score (“POS, grammemes↑”)	F-score (“Categories↑”)	Mistake count
Case=Nom	96.96	96.87	416
Case=Gen	98.20	98.15	237
POS=SCONJ	96.30	96.10	49
Animacy=Anim	96.85	97.03	211
Case=Loc	98.44	98.54	65

Table 3: Grammemes with statistically significant difference between best and worst orders on SynTagRus 2.1.

можно наблюдать при шанхайском аэропорте, не кончаются до самого горизонта.» (‘The difference is that household objects which can be seen near Shanghai airport do not end till the very horizon.’), while the “Categories↑” model predicted accusative. This occurred because animacy—which is less frequent and thus predicted first under “Categories↑”—became conditioned on the attractor, leading the model to an incorrect case at the next step. When case is predicted first (as in “POS, Grammemes↑”), this error is avoided (see Figure

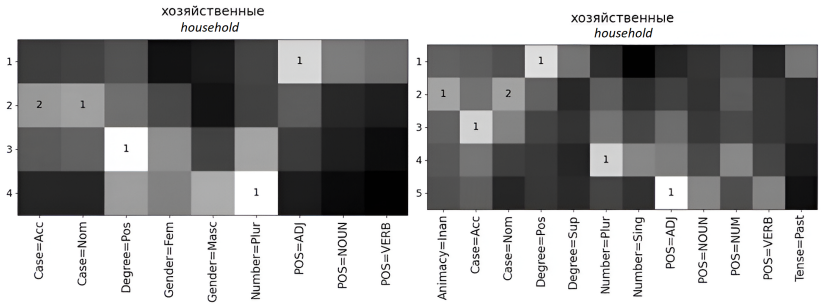

Figure 2: Heatmap of grammeme weights for *хозяйственные* ‘household’.

Similarly, the reverse-order model correctly identified the perfect aspect of *приспособил* ‘adapted’ in the sentence *«Самые первые часы, которые кто-либо приспособил для ношения на руке, неизвестны.»* ‘The very first watch that anyone adapted to wear on their hand is unknown.’. It happened because

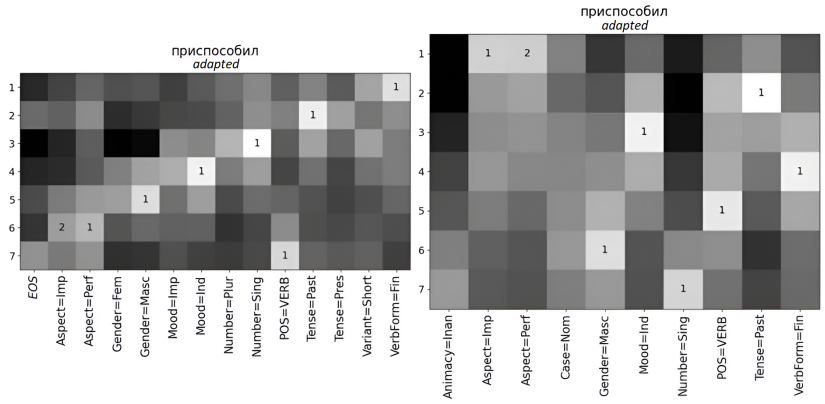

Figure 3: Heatmap of grammeme weights for *приспособил* ‘adapted’.

the model predicted tense first—establishing a context where the perfect aspect is natural in Russian [Paducheva, 2004]—while the “Grammemes↑” model, lacking this context at the aspect-prediction step, made an error.

7 Conclusion

Grammeme prediction order affects the accuracy of sequential neural morphological tagging for Russian, and the difference is statistically significant. The main quality differences concentrate in a small set of difficult grammemes—case, animacy, part of speech, and aspect—where per-grammeme F-score changes can correspond to hundreds of incorrectly tagged tokens despite small macro-level differences. The effect is explained by grammeme dependencies: predicting a grammeme earlier provides useful context for subsequent predictions.

Future work should (1) investigate optimal order selection, either via linguistically motivated orderings or automatic methods such as OAXE, (2) test the findings with modern pretrained encoders such as BERT, and (3) verify whether similar effects occur in other morphologically rich languages.

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